

Pramodit Adhikari

CONTACT INFORMATION

University of Nebraska-Lincoln
Durham School of Architectural Engineering and Construction
Peter Kiewit Institute,
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EDUCATION

University of Nebraska-Lincoln, Omaha, NE

Ph.D., Architectural Engineering, Jan 2024 – Present

- Transferred from Civil and Environmental Engineering (Jan 2021 – Dec 2023)
- Adviser: Professor Milad Roohi
- Area of Study: Structural Engineering

Colorado State University, Fort Collins, CO

M.Sc., Department of Civil and Environmental Engineering, Aug 2020

- **Thesis Topic:** *Life-Cycle Cost and Carbon-Footprint Analysis for Buildings and Communities Subjected to Tornadoes*
- Advisers:
 - Professor Bruce R. Ellingwood
 - Professor Hussam Mahmoud
- Area of Study: Structural Engineering

Tribhuvan University Thapathali Campus, Thapathali, Kathmandu, Nepal

B.S., Civil Engineering, Jan 2017

- Structural specialization (emphasis on earthquakes)

CURRENT ACADEMIC APPOINTMENTS

Graduate Research Assistant, University of Nebraska-Lincoln
Durham School of Architectural Engineering and Construction

Jan 2024 to present

- *Ongoing Research:*
 - Community Resilience against Tornado Hazards
 - Bayesian Fragility Update and System Identification using Sparse Data
 - Investigating Seismic Margin of Safety with Minimum Perturbation

PREVIOUS ACADEMIC APPOINTMENTS

Graduate Research Assistant, University of Nebraska-Lincoln
Department of Civil and Environmental Engineering

Jan 2021 to Aug 2023

- *Responsibilities:*
 - Cement-based electrochemical cells for large-scale energy storage
 - Nondestructive diagnosis and probabilistic prognosis of aging plastic pipe

Graduate Teaching Assistant, University of Nebraska-Lincoln
Department of Civil and Environmental Engineering

Jan 2021 to May 2021

- *Responsibilities:*
 - Instructed students in the use of laboratory equipment for concrete mix design, tensile tests, compression tests, and torsion tests for the laboratory session of the required 3-credit course "CIVE 378 Materials of Construction"

Graduate Research Assistant, Colorado State University
Department of Civil and Environmental Engineering

Jan 2018 to Aug 2020

- *Research:*
 - Developed tornado and hurricane fragility functions for 3 archetypes to be implemented in residential building and community-level economic and carbon-footprint analysis

- Modeled archetypes and residential communities, and created a database for cost and carbon footprint values of building components
- Established a generalized cost and carbon-footprint life-cycle estimation and analysis methodology for examining the benefits of different building practices for residential light-frame wood construction subjected to tornado hazards
- Probabilistic simulation of a tornado passing through the modeled community
- Developed wave/surge and hurricane fragility functions for archetypes to compare different retrofit strategies for a residential community near the shoreline and obtain an optimal strategy for damage mitigation
- **Thesis:** *Life-Cycle Cost and Carbon-Footprint Analysis for Buildings and Communities Subjected to Tornadoes*

PREVIOUS PROFESSIONAL APPOINTMENTS

Wind-driven Fire Reliability Fellow,
Insurance Institute of Business and Home Safety
Wildfire Lab, and SC Wind & Hail Lab

May 2025 to July 2025

- *Responsibilities:*
 - Experimental Design for Fire Resistance of Building Components
 - Collaboratively helped design an experimental methodology in a cone calorimeter to quantify the resistance of wood siding.
 - Surface Temperature Prediction of Wood Siding
 - Ideated a methodology for surface temperature prediction using a filtering algorithm for wood siding samples tested in a cone calorimeter

Internship,
Idaho National Laboratory
Hydrogen and Electrochemistry Department

Oct 2022 to Dec 2023

- *Responsibilities:*
 - Proton-electrochemical cells manufacturing and testing platform
 - Collaboratively helped design and implement an electrochemical cell testing platform to enhance research capabilities and testing throughput.
 - Sealing Research and Development
 - Conducted analysis and redesign of sealant configurations, to improve the efficiency of electrochemical cells by optimizing material and durability.

RESEARCH INTERESTS

Bayesian statistics; community and infrastructure resilience; system reliability; natural hazards; vulnerability analysis; energy and sustainability; game theory; resource allocation; machine learning; engineering education

REFEREED JOURNAL PUBLICATIONS

- [1] Adhikari, P., Roohi, M., & Hernandez, E. "Minimal perturbation of seismic records - Application to nonlinear time history analysis of building structures." *Engineering Structures*, **Under review**.
- [2] Adhikari, P., Roohi, M., Wood, R.L., & Roueche, D. "Assessing Predictive Accuracy of Tornado Resilience Models Through Post-event Reconnaissance Data Analysis." *ASCE Natural Hazard*, **Under review**.
- [3] Adhikari, P., Mahmoud, H., Xie, A., Simonen, K., & Ellingwood, B. Life-cycle cost and carbon footprint analysis for light-framed residential buildings subjected to tornado hazard. *Journal of Building Engineering*, 32, 101657. doi:<https://doi.org/10.1016/j.job.2020.101657>
- [4] Adhikari, P., Abdelhafez, M. A., Dong, Y., Guo, Y., Mahmoud, H. N., & Ellingwood, B. R. Achieving residential coastal communities resilient to tropical cyclones and

climate change. *Frontiers in Built Environment*, 6, 576403.
doi:<https://doi.org/10.3389/fbuil.2020.576403>

- [5] Adhikari, P., Mahmoud, H. N., & Ellingwood, B. R. Life-cycle cost and sustainability analysis of light-frame wood residential communities exposed to tornadoes. *Natural Hazards*, 109, 523-544.
doi:<https://doi.org/10.1007/s11069-021-04847-x>

**CONFERENCE
PROCEEDINGS
(PEER-REVIEWED)**

- [6] Adhikari, P., Mahmoud, H., Xie, A., Simonen, K., & Ellingwood, B. Life-cycle cost and carbon footprint analysis for light-framed residential buildings subjected to tornado hazard. *Residential Building Design and Construction Conference*, State College, PA, USA, March 2020.

**CONFERENCE
ABSTRACTS/
PRESENTATIONS**

- [7] Adhikari, P. & Roohi, M. Field-Calibrated Reliability and Resilience Framework for Tornado Design of Residential Buildings. EMI Conference 2026, University of Boulder, Colorado, Boulder, CO, June 2-6, 2026. **Third Place in Student Paper Competition.**
- [8] Adhikari, P. & Roohi, M. Bayesian Methods for Tornado Resilience Model Updating with Empirical Measurements. EMI Conference 2025, University of California, Irvine, Anaheim Marriott, Anaheim, CA, May 27-30, 2025.
- [9] Adhikari, P. & Roohi, M. Leveraging Machine Learning and AI for Enhanced Tornado Resilience in Community Infrastructure. NHERI Computation Symposium 2025, University of California, Los Angeles, CA, Feb 2025.
- [10] Adhikari, P., Roohi, M., Ghasemi S., Wood, R.L., & Roueche, D.B. Enhancing predictive accuracy in tornado community resilience models through hindcasting and post-event reconnaissance data analysis. Engineering Mechanics Institute Conference 2024, Chicago, IL, May 2024.
- [11] Adhikari, P., Abdelhafez, M. A., Dong, Y., Guo, Y., Mahmoud, H. N., & Ellingwood, B. R. Achieving residential coastal communities resilient to tropical cyclones and climate change. Mini-Symposium on Safety Assessment of Aging Infrastructure: From Data to Decision, Engineering Mechanics Institute Conference, Columbia University, New York, NY, May 2020.

**GRADUATE
SYMPOSIUM**

- [12] Adhikari, P., & Roohi, M. "Model Updating with Empirical Measurements for Tornado Resilience Framework." 2026 COE Graduate Student Symposium, University of Nebraska-Lincoln, Lincoln, NE, Feb 2026
- [13] Adhikari, P., Roueche, D.B., Wood, R.L., & Roohi, M. "Investigating transferability and generalization of tornado resilience models through testbed juxtaposition." 2025 COE Graduate Student Symposium, University of Nebraska-Lincoln, Lincoln, NE, Feb 2025
- [14] Adhikari, P., Roohi, M., Ghasemi S., Wood, R.L., & Roueche, D.B. Enhancing Community Resilience Models against Tornado Hazards through Post-Tornado Reconnaissance Data. 2024 COE Graduate Student Symposium, University of Nebraska-Lincoln, Lincoln, NE, May 2024

**ADVISING AND
MENTORING**

Undergraduate Students

- **Christian Anderson**, August 2025 - Present
 - Developing a Climate-Informed Decision Support Framework for Urban Infrastructure Under Extreme Heat: A City of Lincoln Pilot.
- *Recently Funded. Fall, 2026 - Spring, 2027.*
- **Tyler Reida**, Jan 2025 - Present

- Funded UCARE Project: Creating a Predictive Modelling Methodology for the Recovery of Communities after a Tornadoic Event.
- Tyler presented at the Student Days in Lincoln
- Tyler presented at the 2026 NHERI symposium
- Data collection for recovery analysis at various locations in the 2024 Nebraska and Iowa Tornadoes, paths (major areas included - Elkhorn and Blair, Nebraska, and Minden, Iowa).
- **Geoffrey Sanderson**, Jan 2024 - May 2025
 - Utilizing seismic response from sensors for response estimation of the building.
 - Winner from the College of Engineering at the University of NebraskaLincolns Spring 2025 Research Days poster presentation
- **Jordan Taylor**, Jul 2024 - Nov 2024
 - Data collection efforts at Elkhorn, NE, about the recovery efforts from the Elkhorn tornado in April 2024.

TEACHING EXPERIENCE

University of Nebraska - Lincoln

Graduate Assistant

Spring 2025

- CIVE-859 (Reliability of Structures)
 - Delivered a lecture on Conditional Probabilities and Bayesian Theory.

Graduate Asisstant

Spring 2024

- AREN-8940 (Infrastructure and Community Resilience) - 2024
 - Delivered three lectures on Tornadoes, Hurricanes, and Fragility Analysis - formation of hazards, their impacts, and risk assessment methodologies.

Graduate Asisstant

Spring 2026

- AREN-8940 (Infrastructure and Community Resilience) - 2026
 - Delivered one lectures on Tornadoes, and Fragility Analysis - formation of hazards, their impacts, and risk assessment methodologies.

KEY COMPETENCIES

Analytical

- Probabilistic Analysis
- Hazard and Exposure Modeling
- System Reliability Analysis
- Natural Hazards Vulnerability Analysis
- Community Resilience Analysis

Computer Programming:

-  Python
-  R
-  MATLAB
-  Julia

Computer Softwares

-  AutoCAD
-  Fusion 360
-  Adobe Ilustrator
-  Inkscape
- Finite Element Analysis
 -  SAP 2000

-  ETABS
-  OpenSeesPy

Numerical Analysis:

- MATLAB, R, Maple, Mathematica

Python, MATLAB, and R skill set:

- Linear algebra, Fourier transforms, Monte Carlo analysis, statistics, Bayesian Statistics, visualization

Desktop Editing and Productivity Software:

- VSCode, RStudio
- $\LaTeX$,
- Microsoft Office, LibreOffice, OnlyOffice.com, Google Docs

Operating Systems:

- Microsoft Windows, and Linux (Debian and Fedora family)

AWARDS

University of Nebraska-Lincoln

- **Graduate Research Assistantship**, University of Nebraska-Lincoln, June 2021 - present
- **Structural Engineers Association of Nebraska (SEAO N) Fellowship** Fund, Durham School of Architectural Engineering and Construction, University of Nebraska-Lincoln, 2026-2027 academic year
- **Third Place** in Student Paper Competition, EMI Conference 2026, University of Boulder, Colorado, Boulder, CO, June 2-6, 2026
- **Graduate Student Travel Award** in the amount of \$500 on April 2024
- **Graduate Teaching Assistantship**, University of Nebraska-Lincoln, Jan 2021 - May 2021
- **Graduate Research Assistantship**, Colorado State University, Jan 2018 - August 2021

PROFESSIONAL MEMBERSHIPS

- American Society of Civil Engineers (Student Member)
- Earthquake Engineering Research Institute (Member)

REFERENCES

Professor Milad Roohi (e-mail: milad.roohi@unl.edu)

- Assistant Professor, Durham School of Architectural Engineering and Construction, University of Nebraska-Lincoln

★ *Dr. Roohi is my current Ph.D. supervisor.*

Professor Bruce R. Ellingwood (e-mail: bruce.ellingwood@colostate.edu)

- Senior Research Scientist/Scholar, Colorado State University

★ *Dr. Ellingwood was my Master's adviser.*

Professor Richard Wood (e-mail: rwood@unl.edu)

- Associate Professor, Civil & Environmental Engineering, University of Nebraska-Lincoln

★ *Dr. Wood is co-author in an upcoming publication*

Professor David Roueche (e-mail: db0011@auburn.edu)

- Associate Professor, Civil & Environmental Engineering, Auburn University

★ *Dr. Roueche is co-author in an upcoming publication*

Professor Hussam Mahmoud (e-mail: hussam.mahmoud@vanderbilt.edu)

- Professor, Civil and Environmental Engineering, Vanderbilt University

★ *Dr. Mahmoud was my Master's co-adviser at Colorado State University.*